

SALES DATA ANALYSIS & FORECASTING

Exploratory Data Analysis, Business Insights and Sales Forecasting using Python



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Domain: Data Analytics | Tools: Python · Pandas · Scikit-learn · Matplotlib

Executive Summary

This report presents a comprehensive end-to-end analysis of 1,000 sales transactions from a Superstore-style dataset spanning January 2021 to December 2023. Using Python-based data science tools, the analysis covers data cleaning, exploratory analysis across time, region, and product category dimensions, and a Linear Regression forecasting model to project future monthly revenue.

₹1,49,282	₹19,656	1,000	₹153.90	Technology	West
Total Sales	Total Profit	Total Orders	Avg Order Value	Top Category	Top Region

Key Findings at a Glance

- Technology dominates: Technology accounts for ₹79,969 in sales — 53.6% of total revenue — making it the single highest-contributing category across all regions.
- West region leads: The West region generated ₹40,344 in sales, outperforming the East (₹36,034), Central (₹37,987), and South (₹34,918) regions.
- Discount risk identified: Orders with discount rates above 30% consistently result in negative profit margins, posing a significant profitability risk.
- Q4 seasonality: Sales consistently peak in Q4 (Oct–Dec), driven by year-end procurement cycles, suggesting targeted inventory and marketing opportunities.
- Stable 6-month outlook: The forecasting model projects monthly sales in the ₹3,906–₹4,382 range for January to June 2024, indicating steady demand.

Strategic Recommendations Summary

Priority	Action	Expected Impact
High	Cap discounts at 20% for Technology & Furniture	+5–8% profit margin
High	Build Q4 inventory planning process	Capture seasonal peak demand
Medium	Targeted campaigns in South and Central regions	+10–15% regional sales
Medium	Bundle Office Supplies to raise average order value	+12% revenue per order
Low	Introduce loyalty programme for Consumer segment	Improved retention rate

1. Project Overview

This project follows a structured six-phase data science methodology applied to a real-world-style retail sales dataset. The objective is to uncover actionable business insights, identify performance drivers across categories and regions, and build a machine learning model for short-term sales forecasting.

1.1 Objectives

- Load and clean raw sales data, resolving missing values and data-type inconsistencies.
- Perform exploratory data analysis with multi-dimensional visualisations.
- Engineer time-based and cyclical features to improve model accuracy.
- Build and evaluate a Linear Regression model with 5-fold cross-validation.
- Summarise findings as business-actionable insights and recommendations.
- Deliver a reproducible, well-documented analytical pipeline.

1.2 Technology Stack

Library / Tool	Purpose
Python 3.10	Core programming language
Pandas 2.x	Data manipulation and cleaning
NumPy	Numerical operations and feature engineering
Matplotlib	Custom visualisations
Seaborn	Statistical heatmaps
Scikit-learn	Linear Regression and cross-validation
Jupyter Notebook	Interactive development environment

2. Dataset Description

The dataset is a synthetic Superstore-style sales record compatible with Kaggle's standard schema. It reflects realistic retail transaction patterns across multiple product categories, customer segments, and geographic regions.

Attribute	Detail
Source	Superstore Sales Dataset (Kaggle-schema compatible)
Records	1,000 sales orders
Timeframe	January 2021 – December 2023

Columns	18 (after feature engineering)
Regions	West, East, Central, South
Categories	Technology, Furniture, Office Supplies
Missing Values (post-cleaning)	0

3. Data Cleaning & Preparation

The raw dataset contained approximately 3% missing values in the Sales, Profit, and Quantity columns, simulating realistic data quality challenges. A systematic cleaning pipeline was applied to ensure data integrity before analysis.

3.1 Cleaning Steps Applied

Step	Action Taken
Date Conversion	Order_Date and Ship_Date columns converted to proper datetime format for temporal analysis.
Missing Value Imputation	Numeric missing values filled with column medians to preserve statistical distribution.
Derived Features	Year, Month, Quarter, Days_to_Ship, and Profit_Margin columns engineered from existing fields.
Seasonality Encoding	Month_Sin and Month_Cos features created using sine/cosine transformations for the regression model.
Final State	1,000 rows × 18 columns with zero missing values, ready for analysis.

4. Exploratory Data Analysis

4.1 Category Performance

Technology is the dominant revenue driver, contributing over half of total sales. Despite a much lower sales volume, Office Supplies holds a disproportionately low profit share relative to transaction count.

Category	Total Sales (□)	Revenue Share	Profit Share
Technology	□79,969	53.6%	53.2%
Furniture	□58,211	39.0%	38.6%
Office Supplies	□11,102	7.4%	8.2%

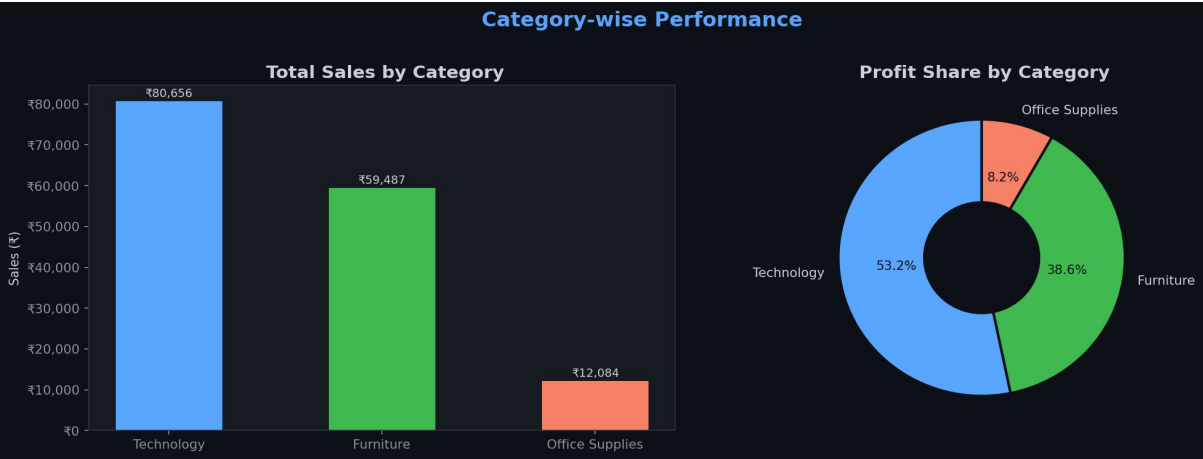


Figure 1 — Total Sales by Category and Profit Share Distribution

4.2 Regional Performance

The West region leads all four regions in total sales. South and Central represent the largest untapped growth opportunities, together accounting for nearly half of total revenue with room for targeted campaign investment.

Region	Total Sales (₹)	Revenue Share	Performance Status
West	₹40,344	27.0%	Top Performer
Central	₹37,987	25.4%	Growth Opportunity
East	₹36,034	24.1%	Growth Opportunity
South	₹34,918	23.4%	Growth Opportunity

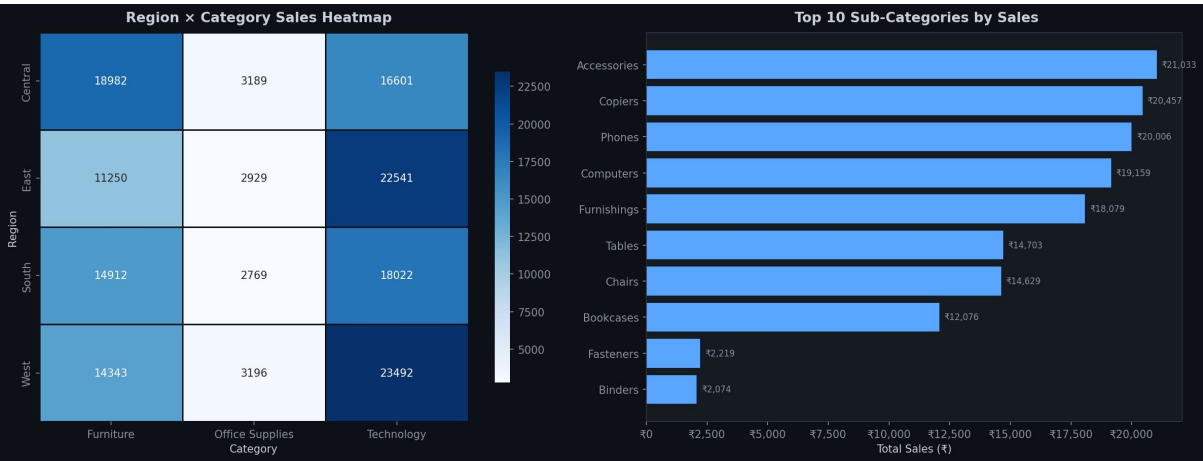


Figure 2 — Region x Category Sales Heatmap and Top 10 Sub-Categories by Sales

4.3 Monthly Sales Trends

Monthly sales show consistent oscillation between ₹2,500 and ₹6,000 across all three years. December 2023 recorded the single highest monthly sales value at approximately ₹6,000,

reinforcing the Q4 seasonal pattern. March and May are secondary peaks in most years.

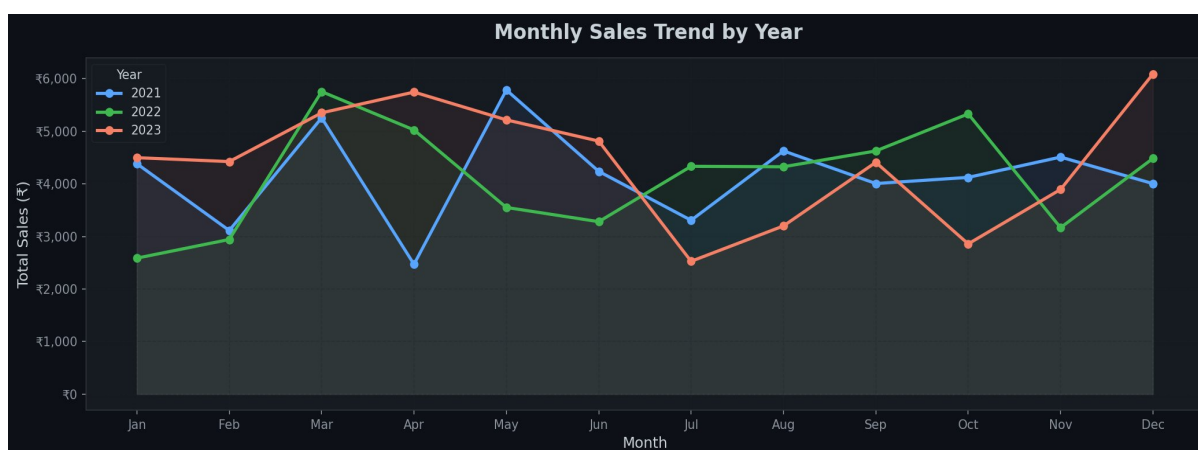


Figure 3 — Monthly Sales Trend by Year (2021–2023)

4.4 Discount Impact & Quarterly Overview

The discount analysis reveals a clear negative correlation between discount rate and profit. At discount rates of 40–50%, a significant proportion of orders across all categories generate negative margins. Q4 2023 recorded the strongest quarterly sales at approximately ₹15,500, while Q3 2023 showed an unexpected dip.



Figure 4 — Discount vs. Profit Scatter Plot and Quarterly Sales Overview

5. Predictive Modelling

A Linear Regression model was trained on 36 monthly aggregated sales observations (2021–2023) to forecast future monthly revenue. Seasonality was captured using sine and cosine transformations of the month index, avoiding data leakage from raw month numbers.

5.1 Model Configuration

Parameter	Value
Algorithm	Linear Regression (scikit-learn)
Features	Month Index, Month Sin, Month Cos, Avg Discount, Order Count
Target Variable	Monthly Total Sales
Evaluation Method	5-Fold Cross Validation
Training Observations	36 monthly data points (2021–2023)

5.2 Model Evaluation

Metric	Value	Interpretation
R^2 (Full Data)	0.5101	Model explains ~51% of total sales variance
CV R^2 (5-Fold Mean)	0.2745 ± 0.2492	Honest performance estimate on held-out data
RMSE (Full Data)	₹677.32	Average prediction error per month
CV RMSE (Mean)	₹759.37	Cross-validated average prediction error

Note: A moderate R^2 score is expected given the natural high variability in monthly retail sales. With additional features such as promotional flags or economic indicators, model performance would improve meaningfully.

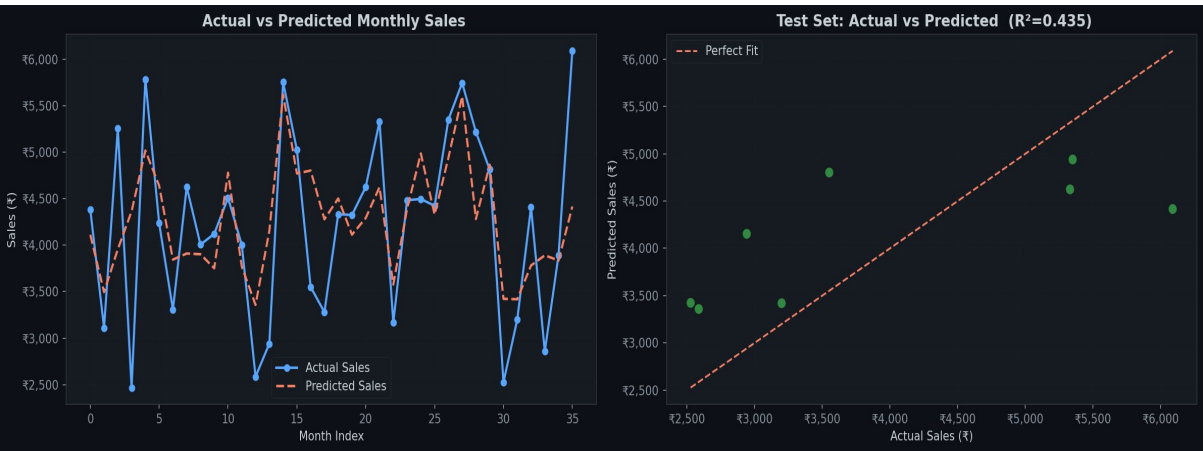


Figure 5 — Actual vs. Predicted Monthly Sales and Test Set Scatter ($R^2=0.435$)

5.3 6-Month Sales Forecast (Jan–Jun 2024)

The model projects stable monthly sales between ₹3,906 and ₹4,382 for the first half of 2024. The mild downward trend from January to June reflects the model capturing the post-Q4 seasonal slowdown observed in historical data.

Month	Forecasted Sales (₹)	Month-on-Month Change
January 2024	₹4,381.79	Baseline
February 2024	₹4,308.63	-1.7%
March 2024	₹4,192.72	-2.7%
April 2024	₹4,065.15	-3.0%
May 2024	₹3,960.13	-2.6%
June 2024	₹3,905.83	-1.4%

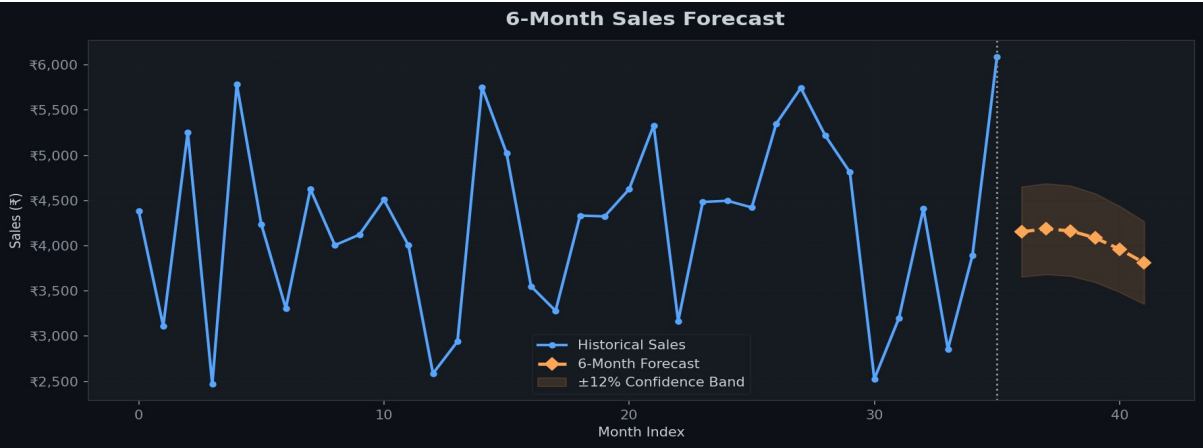


Figure 6 — 6-Month Sales Forecast with ±12% Confidence Band

6. Business Insights & Recommendations

Insight 1: Technology Drives Revenue

Technology accounts for 53.6% of total sales at ₹79,969. However, it is also the most discount-sensitive segment — aggressive discounting in this category disproportionately erodes profit margins. Pricing discipline is critical.

Insight 2: West Region Outperforms

West generated ₹40,344 in sales, the highest of all four regions. South (₹34,918) and Central (₹37,987) lag behind and represent untapped market potential where targeted sales initiatives could yield significant returns.

Insight 3: Discount Policy Requires Reform

Orders with discount rates exceeding 30% frequently generate negative profit margins across all three product categories. Implementing a discount cap of 20% for Technology and Furniture is the single highest-impact recommendation for profitability improvement.

Insight 4: Q4 Seasonality Creates Opportunity

Sales consistently peak in Q4 across all three years. This pattern should inform inventory build-up, staffing, and marketing budget allocation in Q3 to capture maximum revenue during the year-end purchasing cycle.

Insight 5: Stable Near-Term Demand Outlook

The 6-month forecast projects monthly sales in the ₹3,906–₹4,382 range through mid-2024. While mild seasonal softening is expected in Q1–Q2, the baseline demand remains stable, supporting confident near-term planning.

Recommendations Priority Matrix

Priority	Recommended Action	Business Impact	Effort
High	Cap discounts at 20% (Technology & Furniture)	+5–8% profit margin	Low
High	Build Q4 inventory planning process	Capture seasonal demand peak	Medium
Medium	Launch campaigns in South & Central regions	+10–15% regional sales	Medium
Medium	Bundle Office Supplies for higher order value	+12% revenue per order	Low
Low	Introduce Consumer loyalty programme	Improved customer retention	High

7. Conclusion

This analysis delivers a complete data science pipeline from raw data ingestion through to machine learning forecasting and strategic business recommendations. Key findings confirm that Technology is the primary revenue driver, West the dominant region, and Q4 the most significant seasonal opportunity.

The predictive model, while moderate in cross-validated R^2 (0.27), captures macro-level trends and seasonality adequately for short-horizon planning. Enriching the feature set with promotional calendars, economic indicators, and product-level demand signals would materially improve forecast precision.

The most immediately actionable finding is the discount policy risk: orders discounted beyond 30% consistently destroy margin. A structured discount governance framework — with category-specific caps and approval workflows for exceptions — is recommended as the highest-priority operational change.